

Fall 2025, 4100 Problem Set 5

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Let

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & -1 & 3 \\ -6 & 1 & -10 \end{bmatrix}$$

find the LU factorization for A and use it to solve the system

$$\begin{aligned} 2x_1 - x_2 &= 6 \\ 4x_1 - x_2 + 3x_3 &= 1 \\ -6x_1 + x_2 - 10x_3 &= 16 \end{aligned}$$

2. Find the QR factorization of the matrix

$$A = \begin{bmatrix} -1 & 3 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

and use it to find the least squares solution to $A\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

3. Let $A \in \mathbb{R}^{2 \times 2}$ such that $A^2 - A = 6I$ and $|A - 3I| = |A + 2I| = 0$. Compute the determinant of A without using eigenvalues.
4. Let $x_1, x_2, x_3 \in \mathbb{R}$ and denote by $\mathbb{P}_2$ the space of all real polynomials of degree at most 2. Let $T \in \mathcal{L}(\mathbb{P}_2, \mathbb{R}^3)$ be the evaluation operator defined by $T : p(x) \rightarrow (p(x_1), p(x_2), p(x_3))$. Find the matrix representation of T with respect to the standard basis, $\{e_i = x^i : i = 0, 1, 2\}$ and show that its determinant is equal to $(x_3 - x_2)(x_3 - x_1)(x_2 - x_1)$.
5. Let A, B, C, D be $n \times n$ matrices such that A is invertible and consider the block matrix $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$.

- a. Show that the block matrix $E = \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix}$ reduces M to upper-block-triangular form and deduce that $\det(M) = \det(A)\det(D - CA^{-1}B)$
- b. Use the result above to compute the determinant of

$$M = \begin{bmatrix} 1 & 3 & -2 & 1 \\ 4 & 5 & 3 & -2 \\ -1 & 4 & 1 & 3 \\ 2 & -1 & 4 & -5 \end{bmatrix}$$

6. Use Cramer's rule to solve the system

$$\begin{array}{rcl} 2x_1 + 3x_2 + x_3 & = & 8 \\ 4x_1 + 7x_2 + 5x_3 & = & 20 \\ -2x_2 + 2x_3 & = & 0 \end{array}$$